Key Features of Exponential Functions

Name _____

Algebra 2 Lesson 6-1 (fast track) Date _____ Period _____

Learning goal: I can recognize the key features of an exponential function and write an exponential growth or decay model from a starting amount and a percent rate.

Language goal: I can describe an exponential function using the words *growth factor*, *decay factor*, *y-intercept*, *asymptote*, and *end behavior*.

On the quiz: (1) Stretch and reflect 3^x (full Savvas quiz only). (2, bonus) Growth or decay after 5 years, and by about how much? (3) Example 1: domain, range, y-intercept, asymptote. (4) Example 3: write and graph a decay model from a start amount and a percent decrease. (5) Compare growth factors from a graph (full Savvas quiz only).

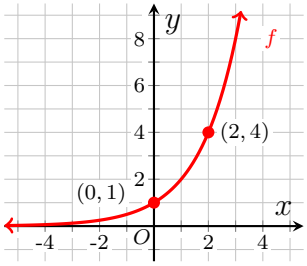
Words: exponential function (*función exponencial*), growth factor (*factor de incremento*), decay factor (*factor de decremento*).

Example 1: Identify Key Features of Exponential Functions

What are the key features of each function? Include domain, range, intercepts, asymptotes, and end behavior. An **exponential function** is any function of the form $f(x) = a \cdot b^x$ where a and b are constants with $a \neq 0$, and $b > 0$, $b \neq 1$.

A. $f(x) = 2^x$

x	$f(x) = 2^x$
-2	$\frac{1}{4}$
-1	$\frac{1}{2}$
0	1
1	2
2	4



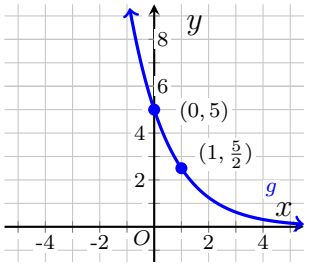
each row: $\times 2$

$b = 2$. Since $b > 1$, the y-values **increase**: 2 is the **growth factor**.

- Domain: all real numbers
- Range: $\{y \mid y > 0\}$
- y-intercept: 1 ($a = 1$, because $b^0 = 1$)
- Asymptote: the x-axis ($y = 0$)
- End behavior: as $x \to -\infty$, $y \to 0$; as $x \to \infty$, $y \to \infty$

B. $g(x) = 5\left(\frac{1}{2}\right)^x$

x	$g(x)$
-2	20
-1	10
0	5
1	$\frac{5}{2}$
2	$\frac{5}{4}$



each row: $\times \frac{1}{2}$

$b = \frac{1}{2}$. Since $b < 1$, the y-values **decrease**: $\frac{1}{2}$ is the **decay factor**.

- Domain: all real numbers
- Range: $\{y \mid y > 0\}$
- y-intercept: 5 (the y-intercept is a)
- Asymptote: the x-axis ($y = 0$)
- End behavior: as $x \to -\infty$, $y \to \infty$; as $x \to \infty$, $y \to 0$

Study tip: the ratio between y-values for consecutive integer values of x is b. **Reason:** in Part B, as x increases the fraction gets smaller and smaller. Will it ever reach zero? _____

Say it: For $g(x) = 5\left(\frac{1}{2}\right)^x$, the *decay factor* is _____, so as $x \to \infty$ the y-values _____. The *y-intercept* is _____ and the *asymptote* is _____ because ...

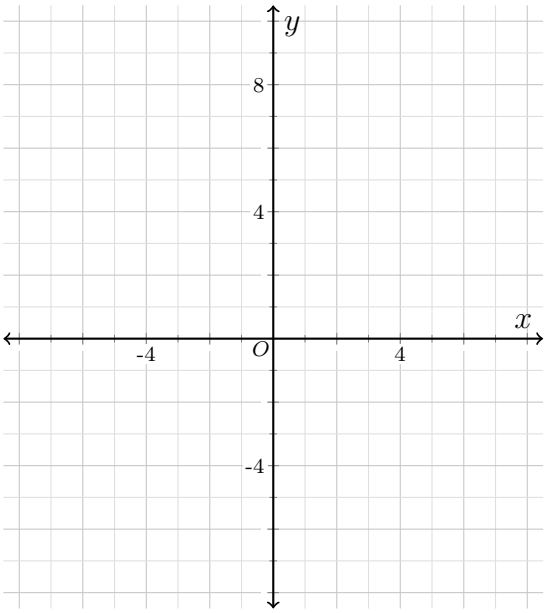
Try It 1: Graph an exponential function

Graph $f(x) = 4(0.5)^x$. What are the domain, range, intercepts, asymptote, and the end behavior for this function?

x	$f(x) = 4(0.5)^x$
-2	
-1	
0	
1	
2	
3	

Growth or decay? $b =$ _____

so the y-values _____.



Domain: _____ Range: _____

y-intercept: _____ Asymptote: _____

End behavior: as $x \to -\infty$, $y \to$ _____; as $x \to \infty$, $y \to$ _____.

How did you know where to start the graph, and which way it goes? Use the words *y-intercept* and *decay factor*.

Example 3: Model with Exponential Functions

The population of a large city was about 4.6 million in the year 2020 and grew at a rate of 1.3% for the next four years.

A. What exponential function models the population of the city over that 4-year period?

Compute the population for the first few years to look for a pattern. Each year you keep all of last year's people and add 1.3% more:

t (years since 2020)	p (population in millions)
0	4.6
1	$4.6 + 4.6(0.013) = 4.6(1.013)$
2	$4.6(1.013)(1.013) = 4.6(1.013)^2$
3	$4.6(1.013)^3$
4	$4.6(1.013)^4$

The 1 in 1.013 is the current population and the 0.013 is the yearly increase. The exponent is the number of years since 2020. The population can be modeled by the exponential function

$$P = 4.6(1.013)^t$$

P: population (millions) after t years 4.6: population in 2020
1.013: growth factor t: years since 2020

B. If the population continues to grow at the same rate, what will the population be in 2050?

2050 is $2050 - 2020 = 30$ years after 2020, so solve the equation for $t = 30$:

$$P = 4.6(1.013)^{30} \approx 4.6(1.4733) \approx 6.78$$

In 2050, the population will be about **6.78 million**.

Exponential growth and decay models. A quantity changes by a fixed percent each time period. Starting amount a, rate r (as a decimal), amount $A(t)$ after t periods:

Growth: $A(t) = a(1 + r)^t$ $a > 0$, growth factor $b = 1 + r$, $b > 1$

Decay: $A(t) = a(1 - r)^t$ $a > 0$, decay factor $b = 1 - r$, $0 < b < 1$

Say it: The *growth factor* is _____ because the city keeps _____% of its people and adds _____%. If a value *dropped* 22% each year, the *decay factor* would be _____ because ...

Try It 3: Write and use a decay model

A factory purchased a 3D printer in 2022 for $35,750. The value of the printer depreciates every year by 22% meaning that it retains 78% of its purchase price for each year that passes.

a. Write the function that represents the value of the printer in terms of x, the number of years since 2022.

$f(x) =$ _____ $\left($_____$\right)^x$ initial value _____ decay factor _____

b. What is the value of the printer after 3 years to the nearest dollar?

c. Does the printer lose more of its value in the first 3 years or in the second 3 years after it was purchased?

Hint: the second 3 years run from year 3 to year 6.

IXL: Practice for the quiz

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Code	Skill	Feeds
LWE	Evaluate exponential functions	warm-up (tables)
EFD ★	Domain and range of exponential functions	Quiz 3
U7P ★	Graph exponential functions	Quiz 4 (graph)
DBW ★	Write exponential functions: word problems	Quiz 4
TYQ ★	Exponential growth and decay: word problems	Quiz 2 (bonus), 4

Exit question: Model with mathematics

A colony of bacteria starts with 50 organisms and quadruples each day. Write a function, $P(t)$, that represents the population of the bacteria after t days. Then find the number of bacteria that will be in the colony after 5 days.

Starting amount (y-intercept): _____ *Growth factor*: _____ because _____

$P(t) =$ _____ After 5 days: _____ bacteria.

Is this exponential growth or decay? What does the *end behavior* as $t \to \infty$ mean for the colony?